

Advances in the study of exosome-derived miRNAs in the pathogenesis, diagnosis, and treatment of systemic lupus erythematosus

Lupus

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Abstract

Systemic lupus erythematosus (SLE) is an inflammatory disease caused by autoantibodies, with high morbidity and mortality. It involves multiple systems, particularly the renal, which can lead to lupus nephritis (LN); its multi-system effects have a significant impact on the physical and mental health of patients. Exosomes are vesicles that are secreted during cell activity and carry a variety of nucleic acids, proteins, and lipids. They are distributed through body fluids for cellular communication. MicroRNAs (miRNAs) are nucleic acids that are packaged within the exosome that are taken up and released in response to changes in plasma membrane structure. MiRNAs are potential participants in immune and inflammatory responses, which are transported to target cells and can inhibit gene expression in receptor cells. It has been suggested that exosomal miRNA can regulate the pathogenesis of SLE and, as such, they are of value in diagnosis and treatment. In this paper, we focus on the research progress into exosomal miRNA in SLE and inspire new directions for SLE related research.

Keywords

Exosomes, microRNAs, systemic lupus erythematosus, diagnostic markers, therapy

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Introduction

Systemic lupus erythematosus (SLE) is an autoimmune disease with complex pathogenesis, resulting from a combination of genetic, immune, sex hormone, infectious, and environmental contributory factors. Disorder of the immune is the key factor of SLE; for example, abnormal activation of macrophages can promote the inflammatory response, thereby aggravating SLE lesions.¹ SLE lesions involving the kidneys develop into lupus nephritis (LN), which is the leading cause of death in SLE patients. The main pathophysiologic mechanism of LN involves the deposition of immune complexes formed by autoantibodies bound to antigens in the glomerulus, which causes glomerular involvement. The studies of epidemiological have revealed that the prevalence of SLE is higher in adults, women, and non-Caucasians.² Currently, clinical manifestations and serological indicators such as anti-double-stranded DNA antibodies and anti-SM antibodies are the main tools for diagnosing SLE. However, SLE is a multi-system disease with a complex and variable clinical presentation that can frustrate an accurate and comprehensive diagnosis. Clinical diagnosis of LN mainly depends on renal

biopsy, an aggressive and expensive procedure. As such, there is an essential need to identify new biomarkers that can be obtained less invasively, and those that are more accurate, and can assist in early diagnosis and accurate prognostication. Corticosteroids and immunosuppressive drugs are the mainstays of treatment for SLE. Although they can alleviate the immune inflammatory response during disease progression, drug toxicity has limited their use.³ Finding new treatments and therapeutic targets to improve the symptoms of SLE is the current research trend.

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The 2013 Nobel Prize in Physiology or Medicine was awarded to the three scientists who discovered and elaborated on the extracellular vesicle transport system and its regulatory mechanisms, making the study of exosomes a worldwide research hotspot. Exosomes, which are widely present in human body fluids, are the most studied and widely used class of extracellular vesicles with cellular communication functions. During biogenesis, exosomes can take up biomolecules such as proteins, lipids, and nucleic acids, and within the protection of a lipid bilayer, these inclusions can be stably transported to the corresponding target cells where they can play a regulatory role. This provides exosomes with the potential to become disease biomarkers, nano-drug carriers, and therapeutic targets.

microRNAs (miRNAs) are non-coding single-stranded RNA molecules that are stably present in exosomes and are targeted for transport to recipient cells to regulate gene expression. miRNAs can regulate immune and inflammatory responses.⁴ In recent years, with the increase of exosomal miRNAs in human disease-related research, our understanding of their role has gradually deepened. Several studies have revealed that exosome-derived miRNAs show utility in the diagnosis, prognosis, and treatment of autoimmune diseases. This review will mainly focus on the application of exosomal miRNAs in SLE.

Introduction to exosomes

Biological characteristics of exosomes

Cellular communication is an efficient and precise response that is an essential part of the body's physiological activity; exosomes play a role in this process as transport regulators. Exosomes originate in the cell nucleus and have a diameter of approximately 40–160 nm (average 100 nm) and can carry nucleic acids, proteins, and lipids. Exosomes play an important role in material transport and cell signaling.⁵

Exosomes can be produced by almost any type of cells and are widely distributed in easily accessible body fluids such as blood, saliva, and urine. The process of exosome production begins with the formation of early-sorting endosomes (ESE). Plasma membrane invagination triggers endocytosis to transport proteins on the cell membrane surface and extracellular protein, ions into the cell. This results in the outward-to-inward outgrowth of the plasma membrane, a process that forms ESEs, which further mature to form late-sorting endosomes (LSEs); a second invagination of the plasma membrane of LSEs forms intraluminal vesicles (ILVs), which are the precursors to extracellularly released exosomes. Multiple ILVs combine to form multivesicular bodies (MVBs), which fuse with autophagosomes and lysosomes for degradation and are transported through the cytoskeleton and microtubule network to the

plasma membrane, where they are released as exosomes with the help of their docking proteins by cytosolic “spitting,”^{4,5} as shown in Figure 1.

Exosomes have a complex composition that includes lipids, proteins, and nucleic acids. These different components play various roles. Lipids form the exosome membrane and are also involved in the process of exosome formation and release. Lysobisphosphatidic acid (LBPA) is a small molecule of glycerophospholipids that accounts for 15% of the total phospholipids present in LSEs; it forms the inner membrane of MVBs and is the key to the maturity of LSEs.⁶ Exosomes are rich in various protein components, such as membrane association proteins that contribute to membrane fusion, tetrapeptides (CD63, CD81, CD9), and heat shock proteins (HSP60 and HSP70); these proteins are exosomal markers. The proteins ARF1, CDC42, and RALA have signal transduction roles.⁷ Exosomes are the link of cell communication by transmitting signals through the intercellular vesicle traffic pathway and fusing with target cells to deliver DNA and RNA from within the exosome to the target cells, thus playing a role in regulating disease processes and immune responses.

Extraction of exosomes

Extracellular vesicles contain exosomes originating from the endosomes of the nucleus and ectosomes produced by exocytosis. These vesicles show similar physicochemical properties to exosomes and thus, the two vesicle classes can be difficult to distinguish. A key technical issue for the accurate study of exosomes and a detailed understanding of their function is their isolation and purification.⁸ Six separation techniques are currently in use, including ultracentrifugation (UC),⁹ ultrafiltration,¹⁰ polymer precipitation,¹¹ immunoaffinity chromatography,¹² size-exclusion chromatography,¹³ and microfluidic-based methods.⁸ The extraction methods of different sample selection are different. For example, exosomes in cell culture medium and serum usually use UC, commercial kits (ExoQuick or Total Exosome Isolation Reagent (TEI)). TRIzol-LS, HiPure Liquid RNA/miRNA kit, Total Exosome RNA Isolation (TER) and exoRNeasy are common methods for extracting exoRNA. There are differences in the concentration, purity and size of exosomes and exosome RNA extracted by different methods. Among them, serum exosomes extracted from UC have high purity but low efficiency of UC and will destroy the structure of exosomes and damage subsequent analysis.¹⁴ The study found that the combination of TEI and TER methods to obtain RNA from cell culture medium has high extraction efficiency and purity.¹⁴

The extraction of exosomes is influenced by their inherent heterogeneity and various other factors, such as the complexity of extraction and the overlap of biological properties between exosomes and other extracellular vesicles. These factors make improving the yield of exosomes,

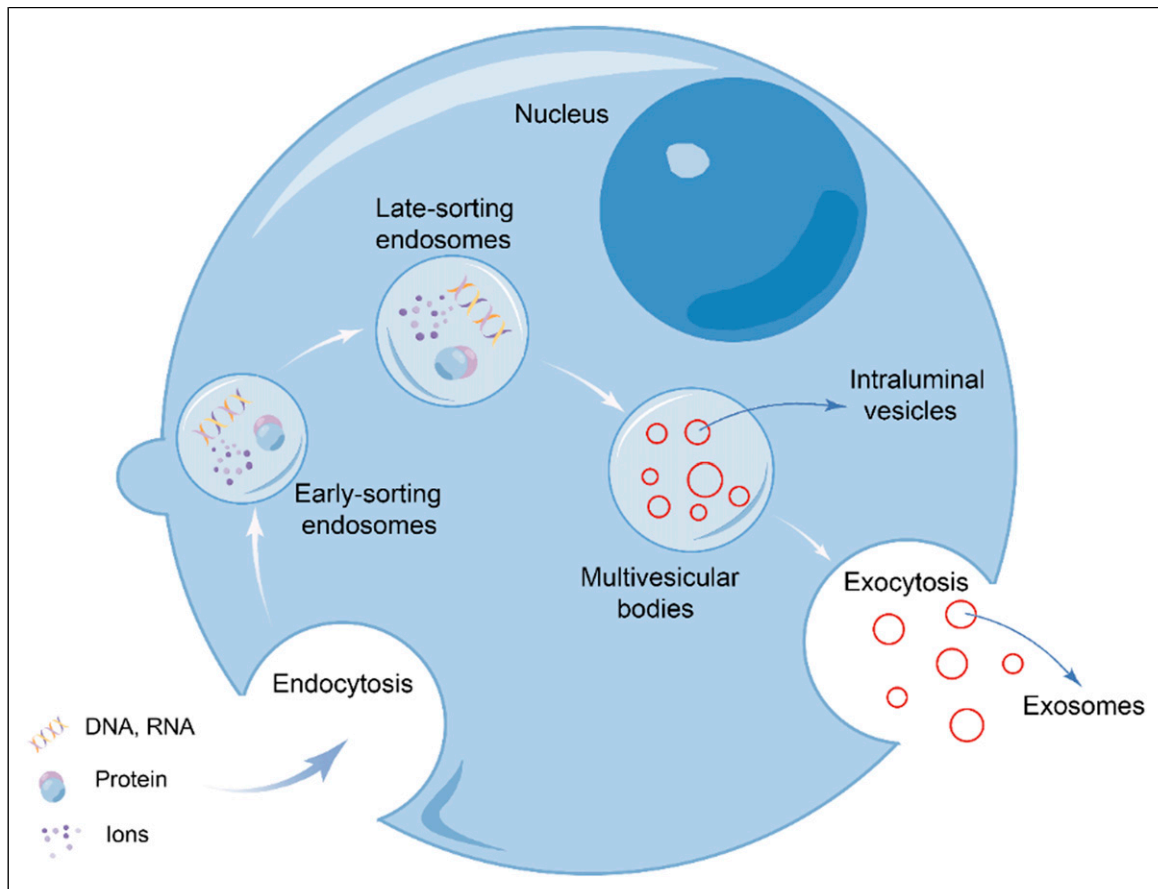


Figure 1. Biogenesis of exosomes. The cell takes up extracellular components such as proteins, lipids, and ions through endocytosis thereby producing an outward and inward proliferation of the plasma membrane, a process that forms Early-sorting endosomes (ESEs), which further mature to form Late-sorting endosomes (LSEs), and LSEs. A second invagination of the plasma membrane forms intraluminal vesicles (ILVs). Multiple ILVs combine to form Multivesicular bodies (MVBs), which fuse with autophagosomes and lysosomes for degradation and are transported through the cytoskeleton and microtubule network to the plasma membrane, where they are released as exosomes by cytokines with the help of their docking proteins.

optimizing the extraction technique, and enhancing the method of differentiating exosomes from other extracellular vesicles a pressing technological hurdle that needs to be addressed. Perhaps the combined use of multiple isolation methods will solve this challenge; the rapid development of microfluidic technology may particularly help to solve these methodological shortcomings.⁸

Exosomes and other diseases

Exosomes have been extensively studied in many diseases, especially tumors, where they can influence tumor growth and metastasis. A study on ovarian cancer found that exosomal miR-205 secreted by ovarian cancer cells is a regulator of the PTEN-AKT pathway and induces angiogenesis thereby promoting tumor growth.¹⁵ Tumor-associated fibroblasts (CAFs), a major constituents of the tumor mesenchyme, are a significant factor in the

development and metastasis of many malignancies. In a study of gastric cancer, gastric CAFs were found to facilitating the migration of gastric cancer cells via the exosomal matrix metalloproteinase 11 (MMP11).¹⁶ In hepatocellular carcinoma (HCC), exosomal miR-25 released from cancer cells promotes hepatocellular carcinogenesis and participates in the development and metastasis of HCC by inhibiting serine/threonine-protein kinase 1 (SIK1) expression and inducing the activation of downstream Wnt/ β -linked proteins.¹⁷ Ji et al.¹⁸ found that exosomes can target homologous cells and transfer miRNA-129-5p to colon cancer cells. Exosomes loaded with miRNA-129-5p had no apparent inhibiting activity on hepatocellular carcinoma and cervical carcinoma; however, they inhibited the proliferation and metastasis of colon cancer cells and promoted apoptosis. Exosomes are readily available in body fluids such as sanguis and urine, and their contents such as miRNAs, have been validated as diagnostic and prognostic

markers in a variety of diseases. In a study of metastatic prostate cancer (mPCa), it was found that a variety of miRNAs (miR-205-5p, miR-125b-5p, miR-148a-3p, miR-183-5p, miR-425-5p) were differentially expressed in exosomes secreted by mPCa cells compared to non-tumor samples using miRNA microarray analysis, suggesting that a differential expression profile of these miRNAs may be a potential biomarker for mPCa.¹⁹

In addition to their use as diagnostic and prognostic markers and their involvement in tumor pathogenesis, exosomal miRNAs have also been studied in the therapeutic context. Exosomal miR-107 reduces the drug resistance of gastric cancer cells by regulating the HMGA2/mTOR/P-gp axis and increases the sensitivity to chemotherapeutic agents, thereby improving the therapeutic effect.²⁰ Exosomes are very stable and can act as natural carriers of miRNAs, delivering them to the corresponding target cells and then exerting their relevant effects. Exosomes released by adipose mesenchymal stem cells can be used to deliver miR-181-5p to hepatic stellate cells (HSC), restraining the STAT3/Bcl-2/Beclin 1 pathway, increasing autophagy, and reducing hepatic fibrogenic degeneration. Overexpressed miRNA-181-5p has an anti-fibrotic function with its ability to selectively transfer miR-181-5p from exosomes to injured hepatocytes.²¹ This provides a novel approach for the therapeutics of liver-related diseases.

miRNAs, SLE, and LN

Several studies have shown that miRNAs are integral part of the pathogenesis of SLE and LN, and has the potential for diagnosis and treatment.

miRNAs and the pathogenesis of SLE and LN

An important component of SLE pathogenesis is the production of large amounts of autoantibodies by overactivated B cells. A 2015 study found that the increased expression of miR-1246 in B cells from SLE patients reduced the expression of early B-cell factor 1 (EBF1), thereby reducing further mobilization of B cells.²² CD4⁺ T cells exhibit a key immunosuppressive role in SLE; one study found that histone deacetylase 1 (HDAC1) exacerbates SLE by inhibiting miR-124 expressed in peripheral blood-derived CD4⁺ T cells of SLE patients and upregulating interferon regulatory factor 1 (IRF1) thereby promoting CD4⁺ T cell activity.²³ B cell activating factor (BAFF), a B cell survival factor, is the cause of B cell overactivation in SLE patients. In one study, the up-regulated miRNA-152-3p in B cells of SLE patients was found to target and inhibit Kruppel-like factor 5 (KLF5) expression, promoting BAFF expression, leading to B-cell mobilization and the production of large amounts of autoantibodies.²⁴ During the pathogenesis of SLE, immune cells can cause kidney damage. Studies have

found that low expression of miR-4512 in monocytes and macrophages of SLE patients can promote the expression of CXCL2 and Toll-like receptor 4 (TLR4) and further activating NF- κ B-mediated chemokine and inflammatory factor secretion. Animal experiments found that the blockade of CXCL2 reduced various symptoms and renal injury in mice.²⁵ Organ damage in SLE is mostly caused by an inflammatory response, and the NF- κ B pathway is a typical pro-inflammatory pathway. Overexpression of miR-101-3p was found to attenuate the inflammatory response of peripheral blood mononuclear cells (PBMC) in SLE by inhibiting MAPK1 expression and blocking the NF- κ B pathway.²⁶ The aberrant proliferation of mesangial cells causes renal damage. miR-155 was found to inhibit chemokine-CXCL13-induced proliferation and transforming growth factor β 1 (TGF- β 1) production in human renal mesangial cells (HRMC) in LN through the down-regulation of the CXCR5-ERK signaling pathway.²⁷

miRNAs and the diagnosis of SLE and LN

microRNAs are not only involved in the pathogenesis of SLE but also show potential in its diagnosis. Abdul-Maksoud et al. found that miR-181a and miR-223 were independently associated with the disease activity index, proteinuria, and serum creatinine in SLE patients. A combined miR-181a and miR-223 criterion was associated with high LN staging and further differentiated LN from non-LN with a diagnostic accuracy of 93.5% (sensitivity and specificity were 97.8% and 91.6%).²⁸ However, in another study, miR-181a expression was found to be low; these differential findings stem from different study sample sizes and participant ethnicities.²⁹ A recent study found that plasma miR-124-3p and miR-377-3p can be combined to diagnose SLE. The areas under the ROC curve (AUCs) for plasma miR-124-3p and miR-377-3p were 0.714 (95% CI, 0.610–0.820, $p < .05$, sensitivity and specificity were 70% and 68.1%) and 0.705 (95% CI, 0.600–0.809, $p < .05$, sensitivity and specificity were 72% and 57.4%), respectively. A diagnosis combining both showed higher AUCs than by using either alone.³⁰ A study by Zeng et al.³¹ found that increased expression of miR-371b-5p and miR-5100 in CD4⁺ T cells and CD8⁺ T cells of SLE patients could distinguish active from inactive disease. MiR-183-5p was identified as a promising biomarker for SLE by Zhou et al. who confirmed that miR-183-5p expression levels were positively correlated with the SLE disease activity index (SLEDAI) and the amount of anti-double-stranded DNA antibodies.³²

miRNAs and SLE and LN treatment

Several studies have identified miRNAs with potential in the treatment of SLE and LN. MiRNA-152-3p may offer a

novel target for anti-inflammatory therapy in SLE. miRNA-152-3p promotes Toll-like receptors in CD4⁺ T cells by targeting DNA methyltransferase 1 (DNMT1) to inhibit the methylation of myeloid differentiation factor 88 (MyD88) (TLR)-mediated inflammatory responses.³³ Overexpression of miR-101-3p in SLE peripheral blood mononuclear cells (PBMCs) inhibits CD4⁺ T cell differentiation to the Th17 lineage by directly targeting HDAC9 and may be a new SLE therapeutic target.³⁴ One study found that miRNA-145 expression decreased with the increase of vascular injury. miR-145 in juvenile LN inhibited platelet-derived growth factor (PDGF)-BB-induced proliferation, migration, and differentiation phenotypic transformation of human vascular smooth muscle cells (HVSMC); miR-145 may be a new target for LN therapy.³⁵ T helper 17 cells (Th17) not only participate in immune response, but also induce inflammatory injury in SLE and LN. A recent study suggested that miR-590-3p can suppress Th17 cells by inhibiting autophagy; increasing miR-590-3p expression can improve lupus nephritis and skin injury in MRL/lpr mice, providing a possible direction for the treatment of SLE symptoms.³⁶ The organ most severely damaged by inflammation in SLE is the kidney. MiR-146a reduces kidney injury in MRL/lpr mice by inhibiting the expression of *TRAF6* and *IRAK1* target genes and contributes to LN treatment.³⁷

At present, a large of studies have confirmed that miRNAs are involved in the pathogenesis of SLE, with diagnostic and therapeutic potential, but the number of miRNAs in the human body is very huge. The current research is only the tip of the iceberg and requires a lot of experimental verification. A summary of miRNAs known to be involved in the development of SLE and LN is shown in Table 1.

Exosome-derived miRNAs and SLE and LN

Exosomal miRNAs are protected by a lipid bilayer and remain stable in blood without being degraded by RNA enzymes. This property makes exosomal miRNAs a key factor in cellular communication. By fusing with membranes, exosomes deliver their encapsulated miRNAs to target cells and thus exert their effects. Exosomal miRNAs are involved in apoptosis, angiogenesis, inflammatory responses, and immune regulation. Their roles in autoimmune disease have been studied.

Exosome-derived miRNAs and the pathogenesis of SLE and LN

Serum exosomes from patients with active SLE can induce an inflammatory immune response. Exosomes released from mesenchymal stem cells (MSCs) can reduce the

proliferation of T and B cells, with immunosuppressive effects. Multiple studies have found that exosomal miRNAs play a key role in the pathogenesis of autoimmune diseases. In rheumatoid arthritis, exosome-derived miR-6089 can regulate LPS/TLR4 and thus mediate the inflammatory response.³⁸

Several studies have found that exosomes are involved in the pathogenesis of SLE. Tan et al.³⁹ observed the delivery of exosome contents into cells by confocal microscopy. After exosomes were co-cultured with CD4⁺ T cells and B cells, the expression of miR-451a was significantly increased in CD4⁺ T cells and B cells. This result indirectly demonstrates that exosomes act as a cell communication molecule, transferring miR-451a to CD4⁺ T and B cells. Senescence of mesenchymal stem cells (MSCs) plays an important role in the development of SLE. Research shows that serum-derived exosomes from SLE patients can enhance the proportion of SA- β -gal-positive senescent cells, disrupt the cytoskeleton, and inhibit growth. Other studies found that exosomes can act as a cellular communication tool to facilitate the senescence of bone marrow MSCs by activating NF- κ B pathway. Exosomal miR-146a negatively regulates the decrepitude of MSCs by regulating *TRAF6*/NF- κ B.⁴⁰ It has also been found that exosomal miR-146a negatively regulates inflammatory responses by inhibiting *IRAK1* and *TRAF6*⁴¹; this is consistent with the mechanisms of non-exosomal signaling events.³⁷ Let-7a and miR-21 are reduced in urinary exosomes from patients with LN during disease flares.⁴² miRNA-let-7a is highly expressed in renal mesangial cells from mice with LN; high expression of let-7a stimulates the production of the inflammatory cytokine IL-6 in mesangial cells.⁴³ miR-21 is a post-transcriptional regulator in the kidney that amplifies the injury response and leads to increased fibrosis.⁴⁴ Therefore, urinary exosome let-7a and miR-21 in SLE patients may be relevant to the pathogenesis of LN.

Exosome-derived miRNAs and the diagnosis of SLE and LN

Exosomes are present in body fluids allowing for their transport over long distances to target cells and regulate cell function. The collection of many body fluids is a rapid, convenient, and minimally invasive method compared to other approaches such as kidney biopsy. As such, exosomal miRNA may be a promising biomarker for minimally invasive screening.

Several studies have identified exosomal miRNAs as having the potential to become biomarkers for SLE. Li et al.,⁴⁵ found that serum exosome miR-155 and miR-21 are increased in SLE patients and that they are even higher in patients with concomitant LN, thus potentially facilitating the clinical differentiation between LN and non-LN cases.

Table 1. MiRNAs involved in the development of SLE and LN.

miRNA	Site	Reference	Relevant targets	Function
miR-1246	Human B cells	[22]	EBF1	Reduced expression of EBF1 and decreased B-cell activation
miR-124	Peripheral blood-derived CD4 ⁺ T cells	[23]	IRF1/CD4 ⁺ T	Down regulation of IRF1 expression reduces the immune activity of CD4 ⁺ T cells
miR-124-3p	Plasma	[30]	NR	A potential biomarker of SLE and LN
miRNA-152-3p	Human B cells/PBMCs	[24, 33]	KLF5/DNMT1	Promotes B cell activation and CD4 ⁺ T cell-mediated inflammatory responses
miR-4512	Monocytes/macrophages	[25]	CXCL2/TLR4	Regulation of chemokine and inflammatory factor expression
miR-101-3p	PBMCs	[26]	MAPK1/NF- κ B	Reduce inflammation in SLE
miR-155	PBMCs	[34]	HDAC9	Inhibits differentiation of CD4 ⁺ T cells into the Th17 lineage
	HRMCs	[27]	CXCL13	Induced proliferation of HRMC and production of TGF- β 1 in LN
miR-181a	Serum	[28]	NR	A potential biomarker of SLE and LN
miR-223	Serum	[28]	NR	A potential biomarker of SLE and LN
miR-377-3p	Plasma	[30]	NR	A potential biomarker of SLE and LN
miR-145	HVSMCs	[35]	PDGF-BB	New therapeutic targets for LN
miR-590-3p	Th17	[36]	Atg7	Th17 cells were inhibited by inhibiting autophagy

PBMCs: peripheral blood mononuclear cells; HRMCs: human renal mesangial cells; HVSMCs: human vascular smooth muscle cells; Th17: T helper 17; NR: not reported; Atg7: Autophagy related 7.

Serum exosome miR-146a expression is reduced in SLE and is inversely related with anti-dsDNA antibody levels and the erythrocyte sedimentation rate (ESR). However, in a study by Perez et al.,⁴⁶ miR-146a expression was discovered to be significantly higher in urinary exosomes from SLE. This difference may stem from the different exosome sample sources. The level of serum exosomal miR-451a in SLE was negatively related to the SLEDAI score. Serum exosomal miR-451a levels in SLE without renal impairment was remarkable lower as compared with the renal impairment. The accuracy of miR-451a in predicting renal damage in SLE patients was determined using a ROC curve analysis, and AUC value of ROC was 0.801 (sensitivity and specificity were 70% and 100%). Therefore, exosomal miR-451a can be used to assess SLE activity and detect the risk of kidney damage.³⁹ Urine is an ideal biofluid for predicting LN activity. In 2015 Cristina et al.⁴⁷ noted that miR-29c levels in urinary exosomes of LN can early evaluate renal fibrosis and the degree of glomerulosclerosis increases when miR-29c is lowly expressed. As a potential less invasive biomarker, miR-29c levels predicted the severity of LN with the corresponding AUC of 0.946 and high sensitivity and specificity (94% and 82%).⁴⁷ Urinary exosome miRNA expression profiles can predict the therapeutic response in LN. miR-31, miR-107, and miR-135b-5p levels were elevated in urinary exosomes and renal tissues in the post-treatment response group compared to the post-treatment non-response group; the levels of all three miRs can be used as early biomarkers of LN prognosis. Among them, miR-135b-5p was the best predictor of

treatment response in LN cases (AUC = 0.783, sensitivity and specificity were 77.8% and 71.4%).⁴⁸ Perez et al.⁴⁶ noted that miRNAs from LN patients were predominantly enriched in urinary exosomes and that miRNA-146a had the potential to identify active LN. Their subsequent study further supported this view, finding that urinary exosomal miR-146a was negatively associated with traditional nephritis biomarkers such as circulating C3 and C4 complement components and proteinuria. Additional studies agree with this view and suggested that exosomal miR-146a is a predictor of SLE disease activity and was an independent marker of disease episodes at a 36-month follow-up.⁴¹

The diagnosis of SLE currently relies on the observation of clinical manifestations and autoantibody tests. However, SLE clinical manifestations are heterogeneous and the underlying disease is complex. As such, reliance on isolated disease indices is inadequate for formulating a comprehensive diagnosis and determining prognosis. The definitive diagnosis of LN is primarily contingent on renal biopsy, which is an invasive procedure. As such, we need to identify less invasive and more accurate biomarkers that are amenable to frequent sampling to aid diagnosis and prognosis. Exosomal miRNAs are readily available in easily accessible bodily fluids and play a demonstrated role in immune regulation and inflammatory responses. miRNAs have been validated as diagnostic markers for cancer, and show significant differences in expression in SLE and LN, making them useful as disease biomarkers. However, there are few current relevant studies and their sample sizes are

insufficient to allow for these approaches to be incorporated into the clinic. Further exploration of the diagnostic value of exosomal miRNAs in SLE and LN is needed.

Exosome-derived miRNAs and the treatment of SLE and LN

The therapeutic utility of exosomes as packages of endogenous products and as drug carriers is being actively investigated. The small exosomes are not easily phagocytosed by mononuclear macrophages; their packaged microRNAs can effectively bind to target cell mRNA and inhibit gene expression in recipient cells. Due to their properties, exosomes can cross vessel walls and extracellular matrix, giving them a natural drug delivery advantage and making them a promising vehicle for targeted drugs.⁴

In a study by Tan et al.³⁹ treatment with glucocorticoids or hydroxychloroquine was found to increase the levels of exosomal miR-451a secreted by CD4⁺ T cells from SLE patients, suggesting that miR-451a is associated with disease improvement and may be a relevant target for the treatment of SLE. Inhibition of *HIF1A*, a target gene associated with kidney disease, reduces mesangial proliferation and IL-8, CCL2, and CCL3 production in endothelial cells. Overexpression of urinary exosomes miR-135b-5p, miR-107, and miR-31-5p was found to improve kidney disease by inhibiting *HIF1A*, contributing to the development of new therapeutic approaches.⁴⁸ Exosomes have been validated as drug delivery vehicles in a variety of diseases. For example, exosomes secreted by bone marrow MSCs can target miR-125b to *SIRT7* and downregulate the latter, thereby protecting against myocardial ischemia-reperfusion

injury.⁴⁹ Bone marrow MSCs-derived exosomes can also attenuate cerebral ischemia-reperfusion injury in mice by inhibiting microglial apoptosis or by mediating CDK6 inhibition via exosomal miR-26a-5p.⁵⁰ M1 macrophages with pro-inflammatory effects are one of the causes of SLE inflammatory response. Promoting the polarization of M1 macrophages to anti-inflammatory M2 macrophages helps to improve SLE lesions. Previous studies have confirmed that miR-146a can regulate the metabolic pathway of mouse macrophages.⁵¹ Recent studies have demonstrated the mechanism by which exosomes secreted by human umbilical cord mesenchymal stem cells deliver encapsulated miR-146a-5p to M1 macrophages to alleviate the inflammatory response in lupus mice and promote the polarization of M1 macrophages to M2 macrophages by reducing the expression of NOTCH1, thereby reducing diffuse alveolar hemorrhage (DHA) associated with SLE.⁵² In addition to the above studies, another recent study also found that by polarizing macrophages into M2 type to alleviate SLE symptoms. miR-16 and miR-21 in exosomes secreted by bone marrow mesenchymal stem cells target *PDCD4* and *PTEN*, respectively, to promote macrophage M2 polarization and exert anti-inflammatory effects to alleviate lupus symptoms in mice.⁵³ One study used an indirect co-culture model to show that miR-let-7c from MSCs delivered by exosomes into mouse renal epithelial cells, increased miR-let-7c levels and inhibited the expression of fibrotic genes type IV collagen $\alpha 1$ and α -SMA in TGF- $\beta 1$ -stimulated renal tubular epithelial cells to attenuate renal fibrosis.⁵⁴ In addition, exosomal miR-86-5p derived from MSCs can restrain the expression of type III collagen, fibronectin, and α -SMA to reduce renal fibrosis.⁵⁵

Table 2. Exosome-derived miRNAs involved in the development of SLE and LN.

Exosomal miRNA	Site	Reference	Relevant targets	Function
miR-451a	Serum	[39]	CD4 ⁺ T	Potential biomarkers and therapeutic targets for SLE
miR-146a	Serum	[40]	TRAF6/NF- κ B	To determine the potential biomarkers of SLE activity and renal damage
	Urine	[41]	IRAK1/TRAF6	The senescence of MSCs was inhibited
let-7a	Urine	[42]	NR	Negatively regulates inflammatory response
miR-21	Serum	[45]	NR	Independent markers of onset at 36-month follow-up
	BM-MSCs	[53]	PTEN	Distinguish between active and inactive LN
miR-155	Serum	[45]	NR	It helps to distinguish LN from non-LN
miR-29c	Urine	[47]	NR	Promote M2 macrophage polarization
miR-135b-5p	Urine	[48]	NR	It helps to distinguish LN from non-LN
				Novel biomarkers for early renal fibrosis progression in patients with LN
miR-31	Urine	[48]	<i>HIF1A</i>	The best predictor for identifying response to treatment in patients with LN
miR-107	Urine	[48]	<i>HIF1A</i>	Novel diagnostic markers and therapeutic targets for SLE
miR-146a-5p	HUCMSC	[52]	NOTCH1	Novel diagnostic markers and therapeutic targets for SLE
miR-16	BM-MSCs	[53]	PDCD4	Inhibiting inflammatory response, promote M2 macrophage polarization
				Promote M2 macrophage polarization

BM-MSCs: bone marrow-derived mesenchymal stem cells; HUCMSC: human umbilical cord mesenchymal stem cells; NR: not reported.

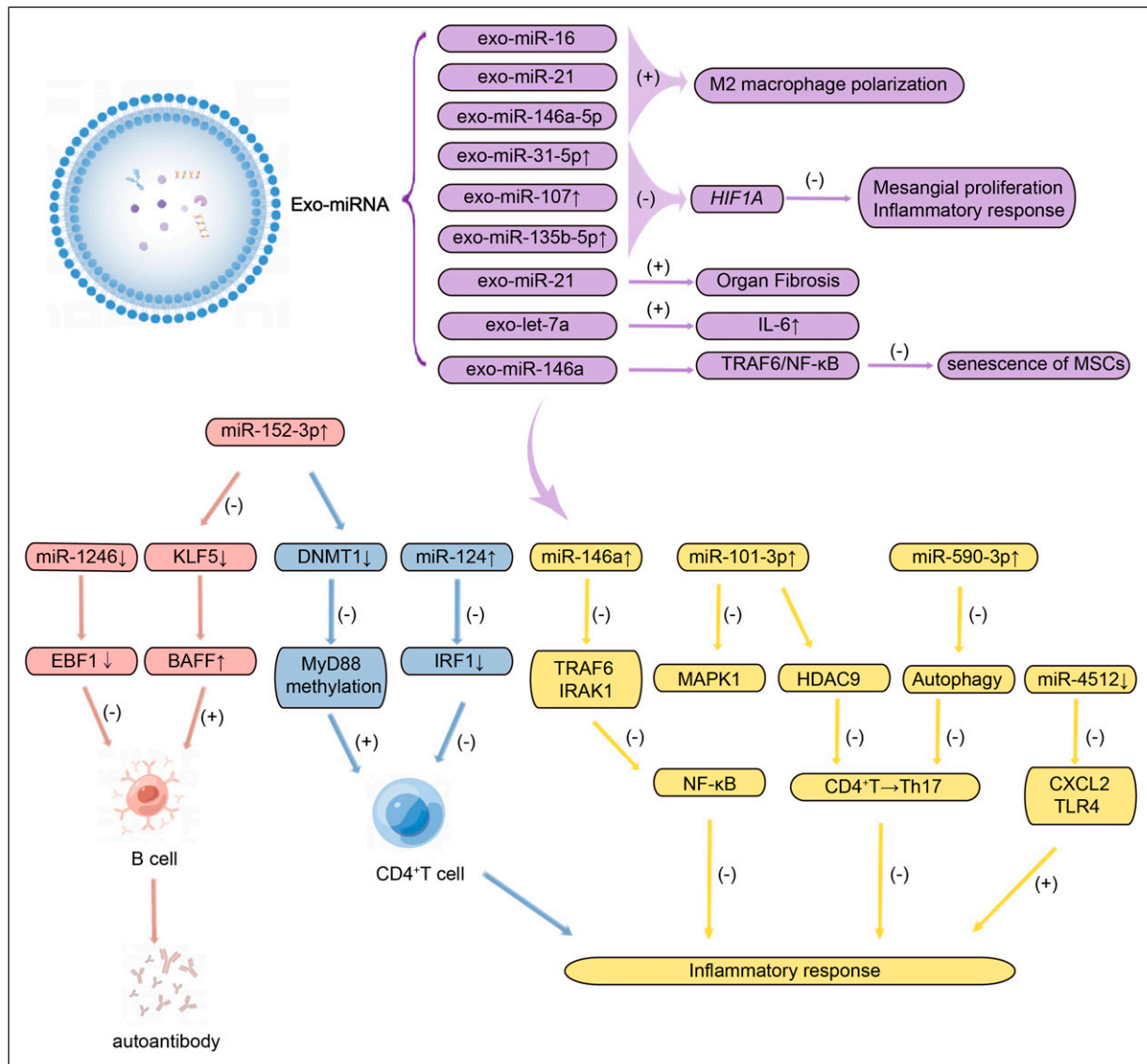


Figure 2. miRNAs involved in the pathogenesis of SLE. Down-expression of miR-1246 reduces EBF1 expression and thus inhibits B-cell activation. Overexpression of miR-152-3p causes an abnormal increase in BAFF expression through inhibition of KLF5 expression, leading to B-cell overactivation. Overexpression of miR-152-3p inhibits MyD88 methylation by reducing DNMT1 expression and thereby promotes CD4⁺ T cell-mediated inflammatory responses. Increased expression of miR-124 inhibits IRF1 expression and thus suppresses CD4⁺ T cell-mediated inflammatory responses. High expression of miR-146a inhibits the classical pro-inflammatory pathway NF-κB by suppressing TRAF6 and IRAK1. Overexpression of miR-101-3p can suppress inflammatory responses either by inhibiting MAPK1 and NF-κB, or by directly targeting HDAC9 to inhibit CD4⁺ T cell differentiation to Th17. Overexpression of miR-590-3p inhibits the differentiation of CD4⁺ T cells into Th17 cells by suppressing autophagy, thereby suppressing the inflammatory response. Down-expression of miR-4512 promotes inflammatory responses by inhibiting CXCL2 and TLR4. High expression of exo-miR-31-5p, exo-miR-107, and exo-miR-135b-5p reduces mesangial proliferative and inflammatory response by inhibiting HIF1A. Exo-miR-21 can promote organ fibrosis. Exo-let-7a promotes production of the inflammatory factor IL-6. High expression of exo-miR-146a also suppresses NF-κB and also inhibits senescence in MSCs via the TRAF6/NF-κB pathway. Exosomal miR-146a-5p, miR-21, miR-16 inhibits inflammatory response by promoting M2 macrophage polarization. ↑: Up-expression, ↓: Down-expression, (+): Promotion, (-): Inhibition, Exo-miRNA: exosome-derived miRNAs.

Current treatment of SLE and LN relies on corticosteroids and immunosuppressants, but these are limited by serious side effects, such as immunosuppression and long-term toxicity. The study of exosomes with lipid bilayers as

drug carriers offers a new elicitation for the therapy of SLE and LN. The studies described in this review have demonstrated that exosomes derived from autologous cells have the preponderance of low immunogenicity, the ability to

cross the blood–brain barrier, and stable long-distance transport of miRNAs, making them favorable drug carriers for the treatment of autoimmune diseases, cancer, and cardiovascular diseases. A summary of the exosome-derived miRNAs participated in the development of SLE and LN is shown in Table 2.

Conclusion

As a correspondent in cell communication, miRNAs are actively involved in the regulation of SLE disease development (Figure 2). The unique advantages of exosomes make them an important mediator of cellular communication and allow them to play a transport role in disease diagnosis and drug delivery. Compared with miRNAs in body fluids, the protective effect of exosomes makes them less susceptible to RNA enzymes and maintains stable expression, which can better reflect the development of the disease. Currently, miRNAs are the most commonly used and tissue-specific biomarkers compared to other nucleic acids in exosomes, for example, miR-122 is mainly expressed in the liver and is almost not expressed in other tissues.⁵⁶ Exosomes are secreted by cells and released into the extracellular fluid, both blood and urine, which are readily available. Urinary exosomes can reflect lesions in the kidney and offer a convenient biofluid sample source for analyzing the diagnostic markers of LN. Samples such as urine are more likely to be accepted by patients because of the relative ease of collection and less invasive nature, compared to histopathological tests such as renal biopsies. Exosomes can deliver their cargo and signal to target cells and regulate their biological functions. They can serve as medication carriers for targeting miRNAs with therapeutic functions to recipient cells and inhibiting the expression of relevant genes. In studies related to SLE, some exosomal miRNAs have shown great potential for diagnosis and prognosis of diseases. These markers must continue to expand related research if they are to be used in clinical practice.

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